
MODULE *fact*

EXTENDS *Naturals*

CONSTANT N

ASSUME $N \in \text{Nat} \wedge N \geq 1$

VARIABLES res, x

$Init \triangleq \wedge res = 1$
 $\wedge x = [i \in 1 \dots N \mapsto \text{TRUE}]$

$OneMult(n) \triangleq \wedge x[n]$
 $\wedge x' = [x \text{ EXCEPT } ![n] = \text{FALSE}]$
 $\wedge res' = res * n$

$Next \triangleq \exists n \in 1 \dots N : OneMult(n)$

$Spec \triangleq Init \wedge \Box [Next]_{\langle res, x \rangle}$
